

SYNOPSIS

Title of the project:

EcoDev: Green Code Analyzer and Laptop Energy Saver

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Project overview:

EcoDev is a local-first software tool that helps developers understand the performance, memory, reliability, scalability, security, and energy implications of their code. It combines code analysis, laptop energy-saving guidance, capacity forecasting, and a clear path toward VS Code and desktop companion integration.

Purpose of the project: (write in points)

- To identify time complexity and memory usage patterns in JavaScript, TypeScript, Python, Go, and C programs, then explain practical alternatives.
- To support user-defined priorities such as fastest runtime, limited memory, reliable and testable code, scalability, security, and bug-resistant design.
- To connect software efficiency with greener computing by reducing unnecessary CPU work, memory movement, background activity, and repeated processing.
- To forecast laptop and resource usage from recent samples so developers can spot rising capacity demand before it becomes a performance or energy problem.
- To encourage everyday laptop energy habits such as automatic screen-off, lower brightness, battery saver, reduced background applications, and keyboard-backlight control.
- To keep analysis local by default, store results in the browser, and explain privacy limits instead of sending source code to a remote service.
- To provide a clear future integration path for a permission-based desktop companion and a VS Code extension that can share active-file and compile-event context locally.
- To help engineers choose an optimization by showing tradeoffs among speed, RAM use, reliability, scalability, security, maintainability, and energy impact.

PROJECT DETAILS

Expected outcomes

- A simple developer-facing interface for comparing code choices with evidence rather than guesswork.
- A more complete view of efficiency that includes code, browser activity, laptop power habits, and workload forecasts.
- Recommendations that are advisory and transparent about confidence; EcoDev does not claim compiler-grade, security-certified, or bug-free results.
- A foundation for future local integrations without requiring accounts, cloud source-code uploads, or unnecessary data collection.

Important implementation note

A web browser cannot directly enumerate every laptop process, change hardware brightness, turn off the operating-system display, or switch the system battery mode. EcoDev therefore presents these as clear local recommendations and keeps the actual device-control path reserved for an optional desktop companion with explicit user permission.

Bibliography / Sources

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